

Mechanical Engineering Portfolio

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Turner Bumbary | Overview

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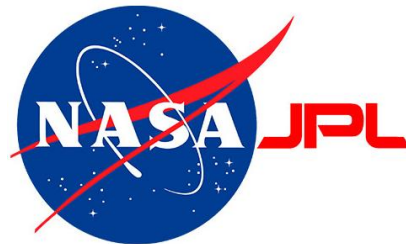
Turner Bumbary | About Me

I am a third-year student at Caltech studying Mechanical Engineering. My professional experiences include developing scientific instruments at NASA and supporting military space acquisitions. Through my role at your organization, I hope to draw on my experiences from NASA, Caltech, and the government to contribute to your teams/products.

Education

The logo for Caltech, featuring the word "Caltech" in a bold, orange, sans-serif font.

Professional Experience



Summer 2020
Winter 2022 - Present



Summer 2021

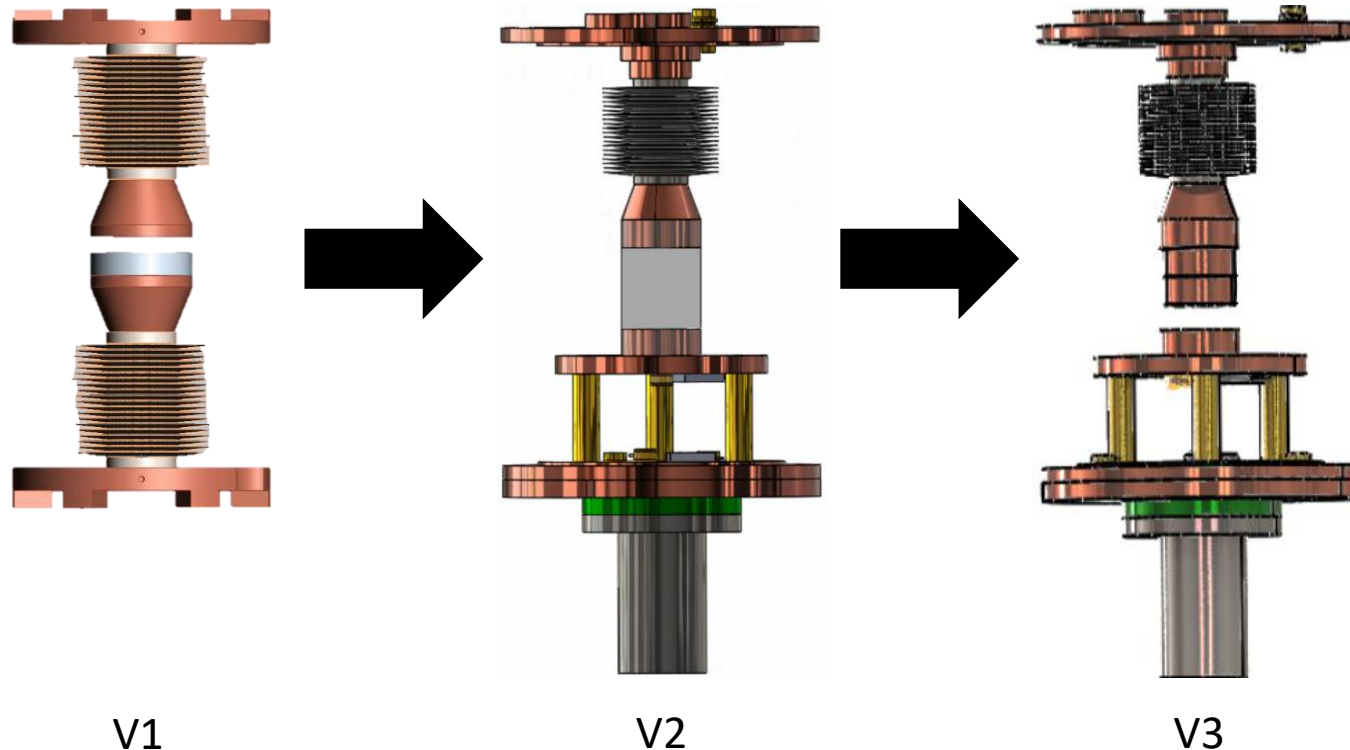


Summer 2022

Turner Bumbary | Scientific Instrument



Purpose: Design a mechanical apparatus to measure the thermal conductivity of lunar soil.

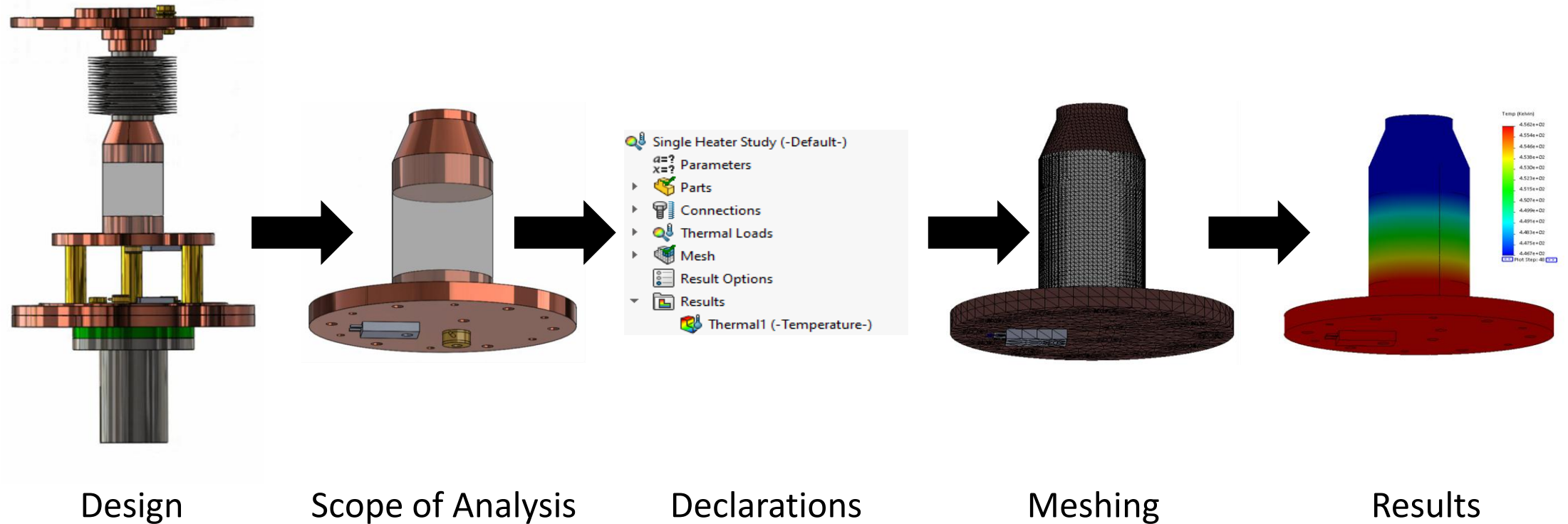


Design: The design utilizes air-filled bellows to compress the lunar soil and metal platens to heat the sample.

Turner Bumbarry | Scientific Instrument



SOLIDWORKS Simulation Workflow: Ran SolidWorks simulations to determine optimal designs/materials of the mechanical device.



Turner Bumbary | Facebook Rooms App



Image courtesy of Meta Inc.

Purpose: Facebook rooms is a platform for users to video chat with their friends/family from the Facebook app.

My Project: I launched the iOS screensharing feature for the platform by integrating the app's engine with the Apple Development Toolkit. This allows users to share content from their devices.

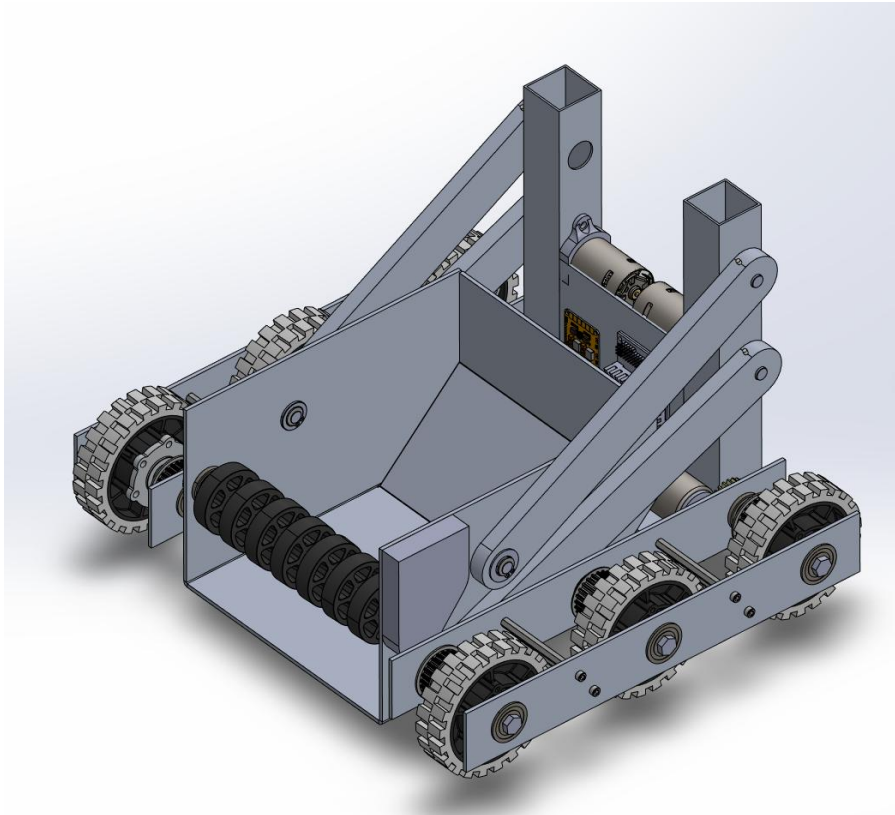
Skills:

- Objective C/C++
- UI/UX Design
- iOS Development
- Front & back-end engineering

Turner Bumbarly | Robotics Competition



Goal: Construct an RC robot capable of collecting racquet balls for a robotics competition.



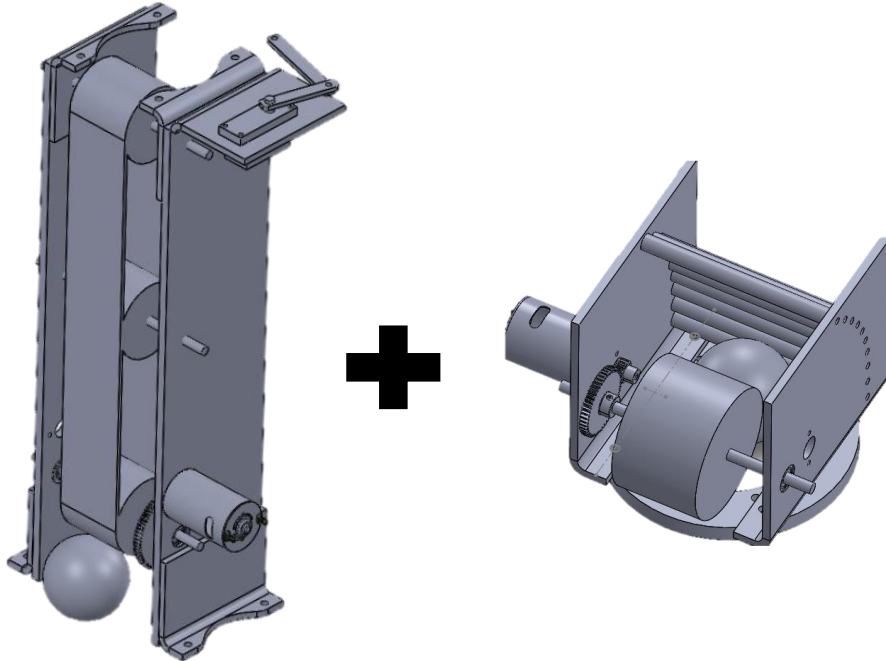
Design:

- Six-wheel belt drive
- Complaint wheels for ball intake mechanism
- Four-bar lift mechanism for collection box

Turner Bumbarly | Robotics Competition



Goal: Construct a robot capable of collecting and shooting racquetballs across the field at 25mph.



Ball-Intake Elevator

Hooded Shooter

Design:

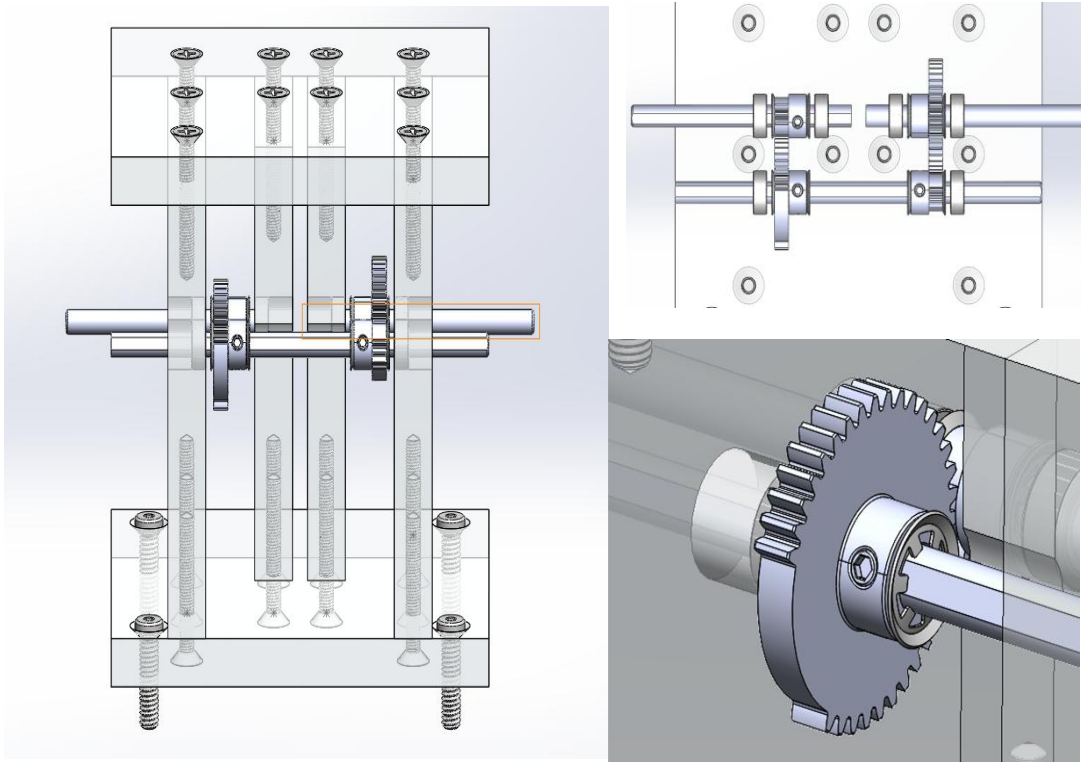
- Ball-intake elevator used to raise balls from the ground intake to the desired clearance height
- Hooded shooter mechanism with spinning flywheel ejects balls at high speeds

Results: Robots passed initial inspection in Dec 2022. Final competition on May 2023.

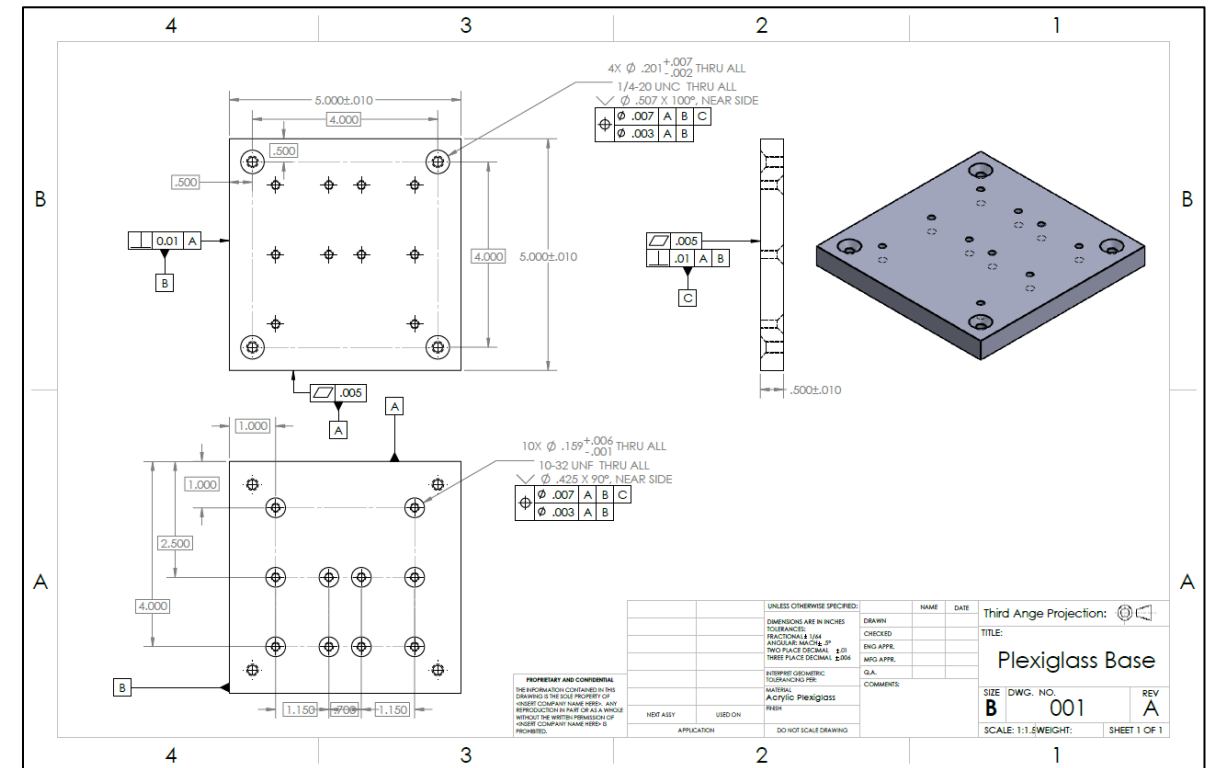
Turner Bumbarly | Transmission Design



Goal: Construct a spur gear transmission to maximize performance of a model drivetrain.



CAD Assembly



GD&T Drawing of Baseplate

Results: Placed first in class competition amongst 30+ students.